VENDORVERIFY AI - COMPLETE BUILD & ARCHITECTURE REPORT

Autonomous AI-Powered Vendor Verification & Compliance Workspace

1. Full-Stack Architecture: React 18 (Vite, TypeScript, Tailwind, Formik, Yup) + FastAPI (Python 3.12).
2. Relational Database: Live MySQL 8 with 10 tables, foreign key cascades, and Alembic migrations.
3. AI Extraction: Dual-engine OpenAI GPT-4o-mini with deterministic local regex/NLP fallback.
4. Deterministic Comparison: 5-way cross-source normalization (MATCH / MISMATCH / MISSING).
5. External Registry: OpenCorporates API integration with resilience & status mapping.
6. Document Checklist & Email: Real-time checklist + automated missing document email dispatch.
7. Verification Workflow: Draft -> Under Review -> Action Required -> Approved / Rejected.
8. Immutable Audit Trail: Version history with JSON diffs and 1-click safe rollback.
9. Compliance Dossier PDF: Dynamic certified compliance dossier export via ReportLab.
10. Discrepancy Resolution: Auditor justification workflows with automated task completion.